

BOEKBESPREKINGS / BOOK REVIEWS

Introduction to Statistics and Computer Programming

by Carl F. Kossack and Claudia I. Henschke

Pilot edition, 651 pages, Holden-day Inc., San Francisco, 98 fig., 5 tables, 60 Fortran programs, \$10-95 (Soft cover)

The authors issue the following challenge to teachers of statistics for second or third year students in the social, biological, management and behavioural sciences:

1. The statistical analyses that will be useful to individuals in the future will differ from the classical techniques.
2. Increased computer capability will enable one to develop techniques which recognize the particular characteristics of the problem being studied instead of using some standard technique which makes assumptions about the problem which cannot be met.

Their response to the challenge is a textbook which emphasizes the role of data analyses in statistics, and links this with decision theory to provide practice in solving practical problems.

Do we believe the authors' view of the future of statistical analyses? Do we want to prepare our students to work creatively on unsolved problems in town planning, transportation, personnel utilization, prevention and treatment of illness? If we believe their view and accept the challenge, we should welcome an indispensable handbook for the task. It uses computer programming to provide a powerful approach to statistical understanding — the simulation of sampling distributions by computer to illustrate abstract theory.

No prerequisites other than high school algebra are required for the students using this text. Instructor's prerequisites, however, include adequate equipment, flexibility in curriculum planning, administrative support, and the imaginative co-operation of their computer colleagues. Joint efforts by instructors of statistics and computing are required to extract benefits for each discipline which this approach can provide. If ability to program in Fortran is not a feasible objective for some groups, the Fortran programs might be rewritten in a less complicated language like Basic. If the objective of any programming capability is rejected, the use of computer simulated distributions could still be achieved by using C.A.I.

In any case, this text is an essential element in rethinking what ought to be done, and a clear example of one way to meet the challenge of a more meaningful and useful statistics course. It covers a wide range of important statistical concepts, including basic description and tabulation, regression and correlation, statistical applications to decision

theory, introduction to experimental design, sample survey design and statistical classification. Every topic is treated in a new way that stresses applications and uses computers to illustrate the theory involved as well as to motivate programming ability and provide experience in creative data analysis.

by J.T. Schneider
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Statistical Quality Control Methods

by Irving W. Burr

Statistics textbook and Monographs Series, Volume 16, Marcel Dekker, Inc., New York and Basel, 522 pages, illustrated, (1976) Price \$29.50.

This textbook, which is suitable at the under-graduate level for statistics and engineering students, is aimed at practical applications rather than complete theoretical development.

The book contains, apart from an introductory chapter, the following:

Some statistical background in Chapters 2 and 15, in which the necessary concepts and formulae on probability theory, distributions and their moments, statistical estimation and hypothesis testing are summarised in 44 pages.

Control charts for measurements and attributes in Chapters 3-8.

Acceptance Sampling for measurements and attributes in Chapters 9-12, including also a summary of the characteristics of various sequential sampling plans.

Some other sampling plans in Chapter 13.

A discussion of the statistics of combinations and tolerances for mating parts in Chapter 14.

The methods discussed are illustrated by real-life examples and problems from practice. In fact the whole approach and discussion suggest that the author has a wealth of experience in this field. Each chapter is concluded with an extensive list of references. The statistics student will find a frustrating lack of theory in Chapter 12 on sequential analysis. This makes it hard to depart from the exact methods given in the book.

The book is neatly typed in double spacing, though paragraph headings could have been more clearly indicated by underlining or italics.

On the whole the book is a valuable addition to a somewhat neglected field as far as textbooks are concerned.

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